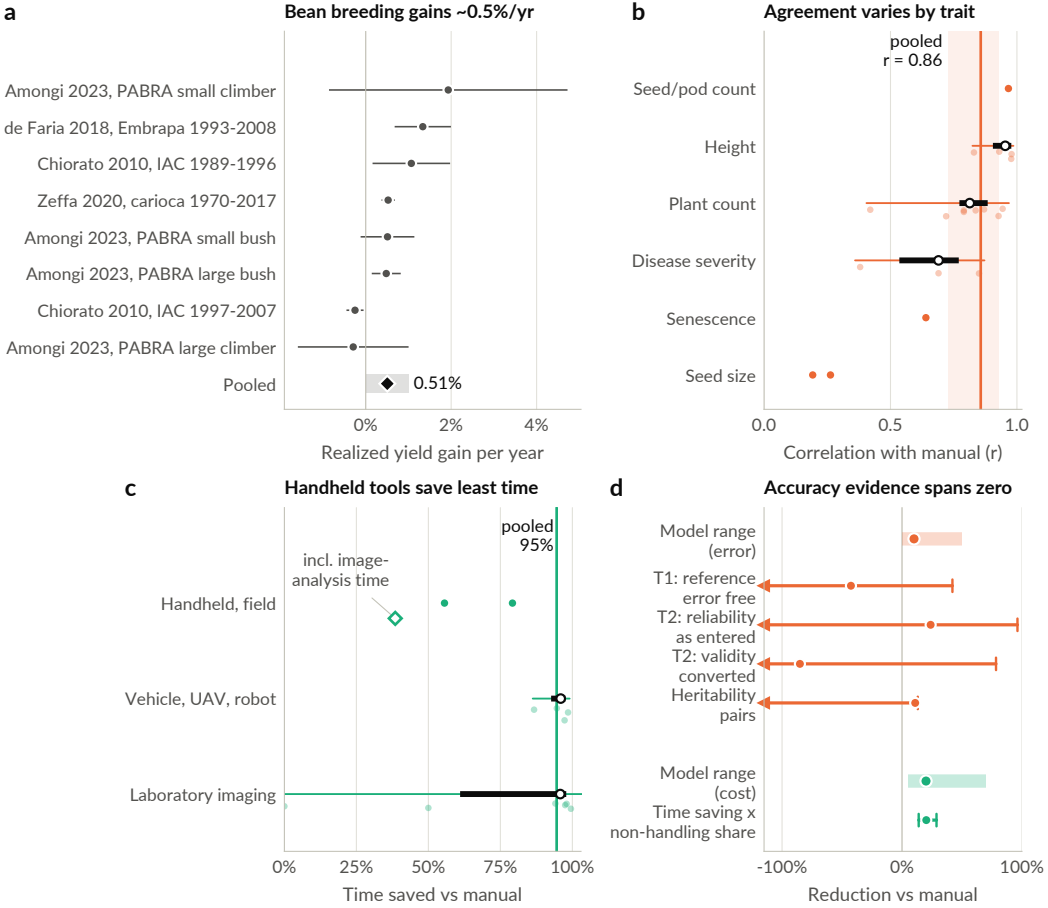
a
Bean breeding gains ~0.5%/yr
Amongi 2023, PABRA small climber
de Faria 2018, Embrapa 1993-2008
Chiorato 2010, IAC 1989-1996
Zeffa 2020, carioca 1970-2017
Amongi 2023, PABRA small bush
Amongi 2023, PABRA large bush
Chiorato 2010, IAC 1997-2007
Amongi 2023, PABRA large climber
Pooled
0.51%
0%
2%
4%
Realized yield gain per year
b
Agreement varies by trait
pooled
r = 0.86
Seed/pod count
Height
Plant count
Disease severity
Senescence
Seed size
0.0
0.5
1.0
Correlation with manual (r)
c
Handheld tools save least time
pooled
95%
incl. image-
analysis time
Handheld, field
Vehicle, UAV, robot
Laboratory imaging
0%
25%
50%
75%
100%
Time saved vs manual
d
Accuracy evidence spans zero
Model range
(error)
T1: reference
error free
T2: reliability
as entered
T2: validity
converted
Heritability
pairs
Model range
(cost)
Time saving x
non-handling share
-100%
0%
100%
Reduction vs manual